



OPEN Cyber resilience across domains: an in-depth exploration of cybersecurity practices and paradigms from the home environment to Operational Technology (OT)

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The intersection of smart home Internet of Things (IoT) devices, enterprise Information Technology (IT) infrastructures and Operational Technology (OT) systems has greatly expanded the cyberattack surface, opening up smart environments to advanced persistent threats (APTs), ransomware, insider attacks and OT protocol exploits. Current cyber security solutions are largely domain specific, and do not have a unified approach for cross domain threat correlation, adaptive response orchestration and proactive cyber resilience. In this regard, this study introduces a novel framework called Cross-Domain Cyber Resilience Framework (CD-CRF) that combines multi-domain telemetry fusion, behavioral anomaly detection, intelligent threat deduction, probabilistic risk scoring, and adaptive response orchestration under a single architecture. The framework utilizes signature independent behavioral analytics and cross domain threat correlation with identification of complex multi-stage attacks in heterogeneous Home, IT and OT environments. Experimental assessment was performed with a set of heterogeneous cybersecurity data sets, including intrusion detection logs, traces of ransomware activities, IoT botnet traffic, SCADA/ICS telemetry, authentication data and communication data for malware. The proposed CD-CRF successfully reduced the false positive rate to 1.8%, and attained a detection accuracy of 98.6%, threat prediction rate of 94.8% and a cross-domain correlation efficiency of 97.1%. Moreover, it had a response time of 120ms, about 50% faster mitigation than traditional methods, and statistically significantly outperformed GCSM and CAT and TinyML models ($p < 0.05$). The results show that CD-CRF improves cyber situational awareness, it increases the ability to detect threats early, it decreases operational risks, and it offers scalable, adaptive cyber resilience solution to converged IT-OT environments.

Keywords Cyber resilience, Cross-domain security, Operational technology security, Behavioral anomaly detection, Advanced persistent threats, Proactive cyber defense

Driven by the rapid development of digital technologies, modern cyber-physical ecosystems have transformed and integrated smart homes, enterprise IT infrastructures, and OT landscapes into connected digital spaces¹. In fact, the Internet of Things (IoT) devices are already at home, and can be used for lighting, HVAC, surveillance cameras and smart appliances automation². The growing trend of distributed cloud platforms, virtualization systems and remote access systems is enabling business organizations to achieve greater operational efficiency³. It is clear that the autonomous operation of industrial systems is controlled by the OTEs in all the sectors of the power and energy, manufacturing, transportation, and utility sectors⁴. Although this digital transformation has brought many advantages in terms of scalability, automation and real-time monitoring, it's also creating more of a cyberattack surface and systemic vulnerabilities⁵.

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Cross-domain cyber resilience

A Cyber Resilience is more than just a Cyber Security; it is the capacity to proactively manage, resist, react and recover from cyber incidents and continue operations⁶. Resilience needs to be pushed in multi-domain systems with various systems of differing priorities and constraints⁷. Besides the convenience and connectivity that is provided by home networks, enterprise IT systems are focused on data integrity and data confidentiality, and OT infrastructures are focused on availability and safety⁸—which is a significant challenge for architectural and operational coordination in protection⁹.

Convergence of home, IT, and OT ecosystems

The days of traditional systems, networks and devices are coming to an end with the rise of cloud integration, the Internet of Things and the remote connection models of home, IT and OT all becoming one¹⁰. Weakened home IoT devices can be a gateway into enterprise networks, and IT system failures can propagate to OT systems, impacting physical operations¹¹. The outcome of this intersection is complex interdependencies that can mean weaknesses in one aspect can be transmitted to other aspects. Therefore, single security measures cannot deal with multi-domain propagation of threats¹².

Escalation of advanced cross-domain threats

Today the cyber adversary uses sophisticated tools such as APTs, ransomware, supply chain attacks and zero-day exploits¹³. Usually these attacks are multi-stage, stealthy, intrusions, which cannot be detected by traditional signature based detection techniques¹⁴. As attackers are able to traverse between the IT and OT networks, operational and financial risks are also heightened¹⁵. Cyberattacks are becoming more automated and sophisticated, and thus requires more intelligent and adaptive defence mechanism¹⁶.

Need for unified deductive and preventive defense

However, even if there are recent developments in the field of IoT security, enterprise defense, and industrial cybersecurity, most of the existing solutions are still disjointed and ineffective¹⁷. Unified threat detection, accurate identification, and proactive prevention mechanisms are urgently needed that can detect behavioral anomalies among the different domains¹⁸. Proposes a CD-CRF that combines multi-layer behavioral analytics, intelligent threat deduction and orchestration of adaptive responses¹⁹. The solution under discussion enhances situational awareness, reduces false alarm rate, and offers a single solution for home, IT and OT interconnected ecosystems to be more proactive when it comes to cyber resilience²⁰.

Contributions

While substantial progress has been made in IoT security, enterprise cybersecurity, and OT protection, most of the solutions proposed so far, such as the Global Cyber Security Model (GCSM), TinyML-based frameworks, and Cybersecurity Awareness and Training (CAT), lack a unified cross-domain cyber resilience. GCSM is mainly concerned with organization awareness and managerial governance, CAT with cybersecurity education and workforce preparedness and TinyML-based methods with lightweight anomaly detection with limited contextual visibility. Such methods are not equipped with built-in multi-domain telemetry fusion, intelligent cross-domain threat correlation, adaptive response orchestration, and inter-connected Home IoT, Enterprise IT, and OT protection capabilities. Therefore, the current study introduces a novel Cross-Domain Cyber Resilience Framework (CD-CRF) consisting of four components: behavioral anomaly detection, probabilistic risk assessment, intelligent threat deduction, and adaptive response orchestration, all integrated into a single architecture that also includes a continuous learning component. One of the biggest innovations of CD-CRF is its ability to conduct signature-independent behavioral analytics and real-time threat correlation across disparate domains, detecting and mitigating multi-stage attacks, advanced persistent threats, ransomware, insider threats and OT-specific exploits. CD-CRF provides proactive defense, automated containment, cross-domain situational awareness with a scalable structure, which substantially enhances detection accuracy, decrease false positive and response time and increase the operational resilience of the enterprise’s cybersecurity beyond the ability of the current cybersecurity model (Table 1).

Related works

In smart homes, IoT enables the home appliances to be connected to manage temperature, detect fires, light the house automatically, and make life more convenient. Such systems also have serious security and privacy concerns including unauthorised access to data and false alarms. It introduces IoT architecture, standards, layered architectures and the key challenges in smart home security, and provides mitigation strategies²¹.

Cybersecurity problems in healthcare are getting worse due to human error and careless cyber hygiene practices. For security, it should be updated regularly and strong passwords should be used and people-

No.	Contribution	Summary
1	Unified cross-domain framework	Integrates Home IoT, IT, and OT into a single, coordinated cyber-resilience architecture
2	Intelligent threat deduction	Uses behavioral modeling and anomaly correlation to detect attacks with precision
3	Proactive adaptive defense	Enables automated prevention and dynamic response across interconnected domains

Table 1. 3 Main contributions.

friendly approaches to education used. There is an emphasis on creating awareness of behavior, best practices in healthcare, and flexible solutions to improve data privacy and security in face of increasing cyberattacks²².

A global cyber security model (GCSM) centered on senior managers is centered on awareness via partnerships, real time threat prevention, and organizational level resiliency. Qualitative interviews and thematic analysis were used to determine key management strategies which included open communication and knowledge sharing. The cooperation with external experts adds to the culture of cybersecurity and strengthens the skills and readiness to existing cyber threats²³.

New cybersecurity standards in Europe and UK for smart homes are based on cloud-based IoT architectures. The benefits of edge computing models, like federated learning and decentralized data storage, overturn these assumptions, offering enhanced privacy and security for edge computing. Home settings have technical and social complexities that highlight the need for flexible standards for external and internal threats²⁴.

Sustainability-oriented and IoT Cybersecurity differ from conventional IT owing to their physical-world interaction and actuation capabilities. Special needs are energy consumption, safety, and privacy. Risk classification, the mitigation purpose, and the problems in implementation are considered, as well as the paradox of openness and tension in sharing data, focusing on the standards development and sustainability-related security case studies²⁵.

The swift digital transformation has increased the cyber threats faced by governments, organizations, and individual people. Some attack vectors, such as malware and denial-of-service attacks, are considered alongside historical and upcoming security frameworks. The strengths, weaknesses, and trends of cybersecurity solutions are analyzed, providing overall insight into evolving threats and defensive strategies²⁶.

More complex cyber threats require new defense models to counter them, rather than the conventional mechanisms. The comparative analysis of the available solutions highlights gaps in their effectiveness and practicality. It presents a new framework for making cyberspace more resilient, which can inform researchers and policymakers. It prioritizes continuous adaptation to address the dynamic and complex digital threat environment²⁷.

IoT devices in smart homes make living easy and more automated, but are highly prone to malware attacks. Viruses take advantage of system vulnerabilities, which are usually embedded in popular applications, and they evolve quickly. The analysis of legal and technical mitigation measures is offered, and it is concluded that the strong security measures are necessary to secure the IoT-enabled smart home space²⁸.

Smart homes are equipped with heterogeneous IoT devices to provide remote control, healthcare, and energy management services. Nevertheless, an internet connection exposes devices to authentication, privacy, and access-control vulnerabilities. Several security measures and authentication techniques are compared, as are countermeasures and open-source research issues aimed at enhancing security in Internet of Things home setups²⁹.

There are integrity-based cyberattacks on AI-powered smart home devices. The anthropocentric interventions in cybersecurity, such as explainable AI and assisted remediation, enable users to detect and mitigate attacks. This is supported by field experimentation showing that multi-modal indicators and empowering users enhance Resilience to AI-oriented cyber threats³⁰.

The development of IoT, cloud computing, and digital services has exposed people to sophisticated cyberattacks like DDoS, phishing, and malware. The classical defense mechanisms are inadequate against evasion. Current technologies, such as machine learning, deep learning, blockchain, and big data, offer the promise of effective detection, but the issue of resistance to adversarial manipulation remains unresolved³¹.

One suggested framework to manage the heightened security risks during the COVID-19 remote work period is the Cybersecurity Awareness and Training (CAT) framework. The framework is divided into three levels and twenty-five core practices aimed at assessing employees' cybersecurity capabilities. Case study validation is effective for identifying skill gaps and enhancing organizational resilience³².

The routing protocol first forms energy-efficient clusters using the Butterfly-Induced Sunflower Optimization (BSFO) algorithm. The Cuckoo Search-based Deep Probability Neural Network (CS-DPNN) selects the optimal Cluster Head (CH) for each cluster based on multiple design metrics. The selected CH acts as an intermediate node, forwarding data between cluster members and neighboring CHs. A Hybrid Recurrent Dynamic Neural Network (RDNN) then chooses the next neighboring CH, enabling reliable and lossless data routing³³.

EMLSCF-FCH (Enhanced Machine Learning-based Secure Communication Framework in Fog Computing-oriented Healthcare), a novel approach to secure, optimize, and enhance disease prediction using ML³⁴.

Tiny Machine Learning (TinyML) is considered a resource-efficient solution for machine learning on more energy-efficient hardware, particularly for cybersecurity. Optimization techniques are used to address problems such as low power consumption and memory usage. A case study experiment on Electric Vehicle Charging Infrastructure demonstrates low delay and memory consumption with small accuracy trade-offs³⁵.

Research challenge: vulnerabilities in smart home, health-care infrastructure, AI-driven devices, and in digital infrastructure. Primarily weak authentication, architecture-based cloud-centric limitations, malware, human error, cyber hygiene, lack of awareness, and new attack strategies that evade traditional protection methods. Current standards and security systems are typically inflexible, non-user-friendly and generally weak against more robust and/or integrity-oriented attacks. This means that interconnected systems will remain exposed to privacy violations, data theft, service disruptions and mass cyber exploitation, as illustrated in Table 2.

The cybersecurity challenges in IoT, healthcare, smart homes, and AI systems require a combined technical and human approach. New structures, edge computing, sophisticated authentication, and machine learning provide a way forward, but dynamic threats require a constant adjustment. Enhancing awareness, norms, and robust architectures is also vital in securing digital ecosystems and guaranteeing long-term security.

The proposed Cross-Domain Cyber Resilience Framework (CD-CRF) shows several fundamental developments over traditional methods of cybersecurity like GCSM, CAT and TinyML. While GCSM centers

Ref.	Domain/environment	Primary objective	Security mechanism	Human-centric support	Key limitation	Performance justification
21	IoT Smart Home	Smart home security and IoT architecture	Layered IoT security framework	No	Limited practical validation	Provides conceptual security guidance but lacks experimental verification in large-scale deployments
22	Healthcare Cybersecurity	Cyber hygiene improvement	Behavioral and educational framework	Yes	Limited technical depth	Enhances user awareness but does not provide automated threat detection capabilities
23	Organizational Security	Enterprise cyber resilience	Global Cyber Security Model (GCSM)	Yes	Qualitative validation only	Strengthens organizational preparedness but lacks real-time threat analytics and automated response
24	Smart Home IoT	Cloud and edge security standardization	Security standards and policy framework	No	Limited empirical deployment	Improves privacy and governance but lacks adaptive cyber defense mechanisms
25	Sustainable IoT Systems	Risk classification and mitigation	Security standards analysis	No	Broad conceptual scope	Identifies security challenges but does not offer implementation-oriented solutions
26	General Cybersecurity	Threat and framework review	Survey-based analysis	No	No unified solution	Provides comprehensive threat overview without a practical detection framework
27	Advanced Cyber Defense	Cyber resilience enhancement	Comparative defense framework	Partial	Limited real-world validation	Introduces resilience concepts but lacks deployment evidence in heterogeneous environments
28	Smart Home IoT	Malware threat mitigation	Legal and technical countermeasures	No	Limited experimental support	Discusses mitigation strategies but lacks intelligent detection capabilities
29	IoT Smart Home	Authentication and access control	Protocol comparison framework	No	Open research challenges remain	Improves authentication but cannot address advanced multi-stage attacks
30	AI-Enabled Smart Home	Explainable cybersecurity	Explainable AI and assisted remediation	Yes	Small-scale validation	Enhances user trust but scalability remains unverified
31	Emerging Cyber Threats	AI-driven attack detection	Machine Learning and Deep Learning models	No	Vulnerable to adversarial attacks	Achieves high detection performance but robustness remains a challenge
32	Remote Work Security	Security awareness and training	CAT Framework	Yes	Limited technical controls	Improves employee readiness but lacks intelligent threat correlation and automated mitigation
35	IoT / EVCI Security	Lightweight cyber defense	TinyML-based anomaly detection	No	Reduced contextual visibility	Efficient for edge devices but cannot perform cross-domain threat correlation
Proposed CD-CRF	Home IoT, IT, and OT	Cross-domain cyber resilience	Telemetry fusion, behavioral analytics, intelligent threat deduction, probabilistic risk scoring, and adaptive response orchestration	Partial	Computational overhead in large-scale deployments	Provides unified cross-domain threat correlation, automated mitigation, continuous learning, high detection accuracy, reduced false positives, and enhanced operational resilience

Table 2. Comparative analysis of existing cybersecurity approaches and research gaps. *GCSM* Global Cyber Security Model, *CAT* Cybersecurity Awareness and Training, *TinyML* Tiny Machine Learning, *CD-CRF* Cross-Domain Cyber Resilience Framework.

on raising awareness within organizations and managerial techniques for coping with threats, the proposed framework offers a full technical architecture that can enable cross-domain threat-correlation across Home IoT, enterprise IT, and OT domains, in real-time. The CD-CRF offers intelligent behavioural analytics, probabilistic risk scoring, and automated adaptive response orchestration, whereas the CAT framework primarily focuses on cybersecurity awareness and employee training for a proactive approach to cyber defence. Unlike the typical TinyML-based approaches which are optimised to detect threats in a lightweight fashion and from limited contextual visibility, the proposed framework is designed to perform unified multi-domain telemetry fusion and coordinated threat inference across heterogeneous environments. The signature-independent anomaly detection, cross-domain behavioural correlation, intelligent threat deduction and adaptive mitigation are all built into a scalable architecture that is a key novelty of CD-CRF. The synergic approach allows early identification of multi-stage attacks, reduction of false positive detection, decrease of response time and increase of operational resilience of interwoven cyber-physical ecosystems.

Current cybersecurity frameworks are primarily designed for different cyber environments, including IoT, Healthcare, Enterprise IT, and OT, with little support for a unified, cross-domain approach to protection. However, while frameworks such as GCSM and CAT are mostly concerned with the awareness and organizational aspects, TinyML and AI-driven models are more targeted at lightweight anomaly detection in a specific domain. These approaches, however, fail to offer a coordinated threat correlation, adaptive response orchestration and scalable protection in multi-stage attacks of interconnected Home, IT and OT ecosystems. By combining behavioral analytics, multi-domain telemetry fusion, intelligent threat deduction, and proactive adaptive response into a coherent cyber resilience framework, the proposed CD-CRF offers finer precision in detection, lesser false alarms, and a better cross-domain security posture.

Proposed cross-domain cyber resilience framework (CD-CRF)

The proposed CD-CRF endows one security architecture capable of stitching together the home security, enterprise IT security and OT security into a unified security environment. The framework will be structured as a multi-layered architecture with four modules related to telemetry fusion, behavioural intelligence, threat deduction, precision identification, proactive response and continuous learning.

Behavioral Intelligence involves the analysis of historical and real-time patterns of user, device, application, network traffic, or industrial processes to create dynamic baseline profiles and determine deviations from them that can be used to detect malicious activity. Intelligent Threat Deduction refers to the ability to identify potential cyber threat, based on the correlation of multi-domain telemetry data, anomaly indicators, contextual information, and probabilistic risk assessment, thereby uncovering hidden relationships in attacks and multi-stage intrusion campaigns that might not be caught by signature-based means. Precision Attack Identification is the ability to accurately identify and validate detected anomalies through risk-scoring, behavioral signatures, threat intelligence and contextual verification technologies to differentiate between true threats and false alarms. Together, they provide a proactive approach to cyber defense by converting raw telemetry data into actionable security intelligence, boosting accuracy in detection and minimizing false-positive alerts, thus helping to build greater cross-domain cyber resilience in Home IoT, Enterprise IT and Operational Technology (OT).

Figure 1 illustrates a single architecture of cyber resilience that combines Home, IT, and OT environments. Multi-domain data acquisition gathers logs, telemetry, and network flows, which are normalized and baselined with behaviour. The system carries out cross-domain correlation, risk detection, and precision threat analysis through signature-independent processes. The elastic prevention and adaptive response layer allows automated containment, segmentation, and automatic policy revisions. Ongoing learning also ensures alignment between IT and OT defenses, enhancing detection accuracy and reducing false alarms across linked infrastructures.

The proposed CD-CRF has been designed with implementation oriented modular components which will help to deploy it in Home, IT and OT environments. The framework leverages standard communication protocols like MQTT, Modbus, TCP/IP, and REST based APIs to collect logs, network flows, authentication records, IoT traffic and OT sensor data using distributed telemetry collectors. Collected data is normalized in a centralized normalization engine and sent to the behavioral analytics and threat deduction modules that are deployed on edge-cloud infrastructure. A machine learning anomaly detection and probabilistic risk-scoring models are combined with automated response orchestration for firewall updates, IoT isolation, network segmentation and privilege control. The modular architecture allows to achieve interoperability, scalability and real-time coordinated defense across the cyber-physical infrastructures composed of heterogeneous infrastructures, which enhances the practical feasibility and deployability of the proposed framework.

Let $D = \{\text{Home}, \text{IT}, \text{OT}\}$ represent the set of operational domains considered in the framework. For each domain $d \in D$, the telemetry feature vector is represented as $X_d(t) \in \mathbb{R}^m$, where m denotes the number of extracted behavioral and network features at time instant t . These features include authentication logs, packet statistics, command sequences, sensor measurements, process telemetry, and IoT communication frequencies.

The baseline behavioral profile of each domain is represented by $B_d(t)$, which is continuously updated using adaptive learning. The parameter $\alpha \in [0,1]$ denotes the adaptive learning rate controlling the sensitivity of baseline updates. In this work, $\alpha = 0.3$ is selected experimentally to balance adaptation speed and behavioral

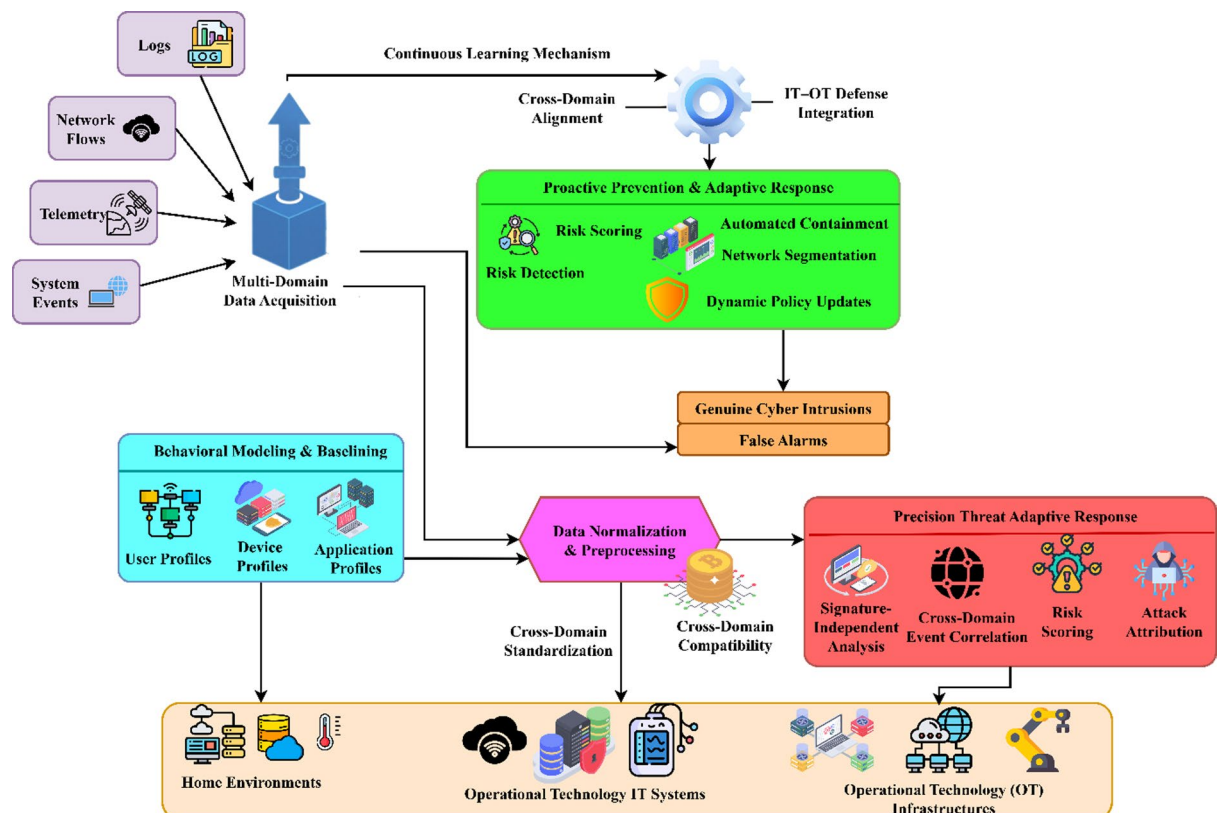


Fig. 1. CD-CRF architecture.

stability. The weighting coefficient ω_d represents the infrastructure criticality factor assigned to each domain, where higher values are allocated to OT systems due to their operational sensitivity.

Cross-domain correlation score C_{xd} computed by Eq. 1

$$C_{xd} = \frac{1}{T} * \text{sim} + (F_d(t), B_d) \quad (1)$$

T denotes the observation window length, while $F_d(t)$ is the multi-modal feature vector extracted at time t from domain d . B_d is the baseline behavioural profile learned for that domain. The function sim efficient w_d weights each domain's importance according to infrastructure criticality.

Signature-independent risk detection $R(t)$ evaluated using Eq. 2

$$R(t) = \sum_d \alpha_d \frac{\|F_d(t) - B_d\|_2}{\sigma_d} + \beta \frac{1}{N} \sum_{i=1}^N A_i(t) \quad (2)$$

$F_d(t)$ and B_d again denote observed and baseline features for the domain d . Their Euclidean distance captures anomaly magnitude, normalized by domain variability σ_d . The weighting α_d adjusts domain contribution. The term aggregates contextual alerts $A_i(t)$ from N correlated indicators, scaled by the coefficient β . $R(t)$ integrates statistical anomaly and contextual evidence to detect cross-infrastructure threats with reduced false alarms.

Elastic prevention and adaptive response effectiveness E_{resp} modelled using Eq. 3

$$E_{resp} = 1 - \frac{1}{M} \sum_{m=1}^M \gamma_m \frac{T_{contain}^{(m)} + T_{recover}^{(m)}}{T_{impact}^{(m)}} \quad (3)$$

For incident m , $T_{contain}^{(m)}$ is automated containment latency, $T_{recover}^{(m)}$ is restoration time, and $T_{impact}^{(m)}$ is the potential impact duration without response. The coefficient γ_m weights incident severity.

Continuous learning alignment across IT-OT-Home L_{align} defined by Eq. 4

$$L_{align} = \frac{1}{D} \sum_{d=1}^D \frac{\langle \theta_d^{(t+1)}, \theta_d^{(t)} \rangle}{\|\theta_d^{(t+1)}\| \|\theta_d^{(t)}\|} \cdot \left(1 - \frac{FP_d + FN_d}{N_d}\right) \quad (4)$$

D is the number of domains $\theta_d^{(t)}$ and $\theta_d^{(t+1)}$ are model parameter vectors before and after learning updates. The detection accuracy FP_d and FN_d are false positives and false negatives in domain d , and N_d is the total number of evaluated events. L_{align} is high when model updates remain consistent while improving detection reliability across interconnected infrastructures.

Multi-domain data acquisition and normalization

This layer gathers heterogeneous data from IoT devices, routers, servers, cloud platforms, firewalls, PLCs, SCADA systems, and industrial sensors. Available data sources include system logs, NetFlow logs, command sequences, authentication logs, and process telemetry. These datasets have different structures and protocols, so a normalization mechanism unifies the attributes, timestamps, and metadata into a single schema. To ensure cross-domain analytics compatibility, noise filtering, feature extraction, and encoding methods are used. A single data foundation enables accurate correlation of Home, IT, and OT ecosystems.

Behavioral modeling and baselining

The framework forms dynamic behavior baselines for users, devices, applications, and industrial processes. The algorithms of statistical and machine learning analyze past activity to determine the usual mode of operation. Profiles are also constantly updated based on acceptable differences in usage and workload. In home networks, the model monitors the frequency and access behavior of IoT communication; in IT systems, it monitors login and log-out patterns; and in OT systems, it monitors control command and sensor timing sequences. These thresholds serve as a reference point for identifying abnormalities that indicate a cyber threat.

Figure 2 illustrates a cross-domain behavioral modelling system of Home Networks, Enterprise IT and OT Infrastructure. It investigates the user login behavior, the frequency of communication and sensor values to IoT devices and sequences of control through statistical profiling and time-series learning. Adaptive baselines represent the normal operation, and allow dynamic anomaly detection. Contextual analysis identifies deviations in data exfiltration, suspicious IoT activity, etc., or refinement of the predictive accuracy of threat detection and resilience building in the networked cyber-physical space in general.

Cross-domain behavioral profile estimation $P_d(t)$ dealt using Eq. 5

$$P_d(t) = \frac{1}{W} \sum_{\tau=t-W+1}^t \phi_d - (U_d(\tau), I_d(\tau), S_d(\tau), C_d(\tau)) \quad (5)$$

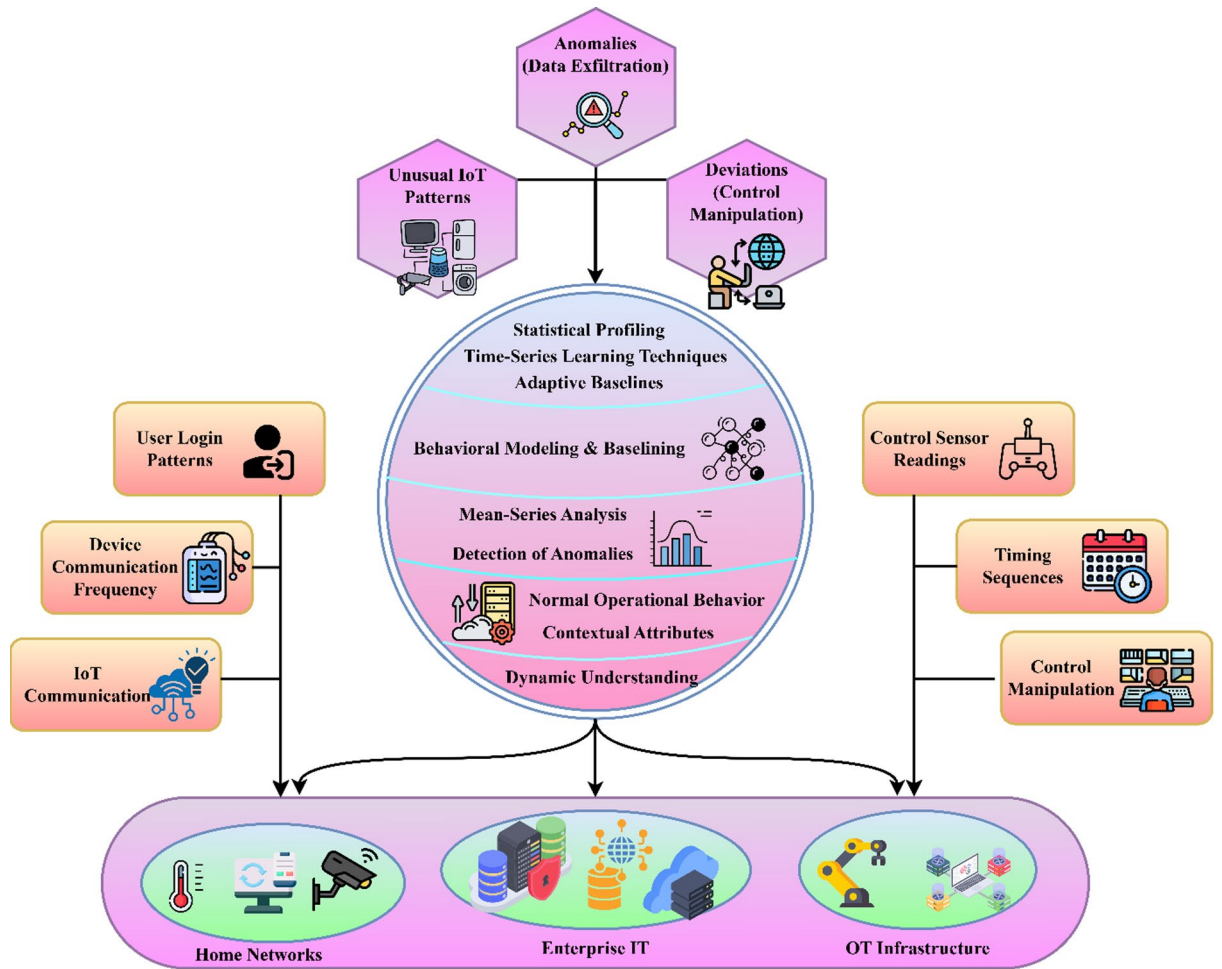


Fig. 2. Behavioral modeling and adaptive anomaly detection across domains.

W is the sliding observation window capturing recent behaviour. The function ϕ_d extracts statistical-temporal features from four behavioural streams: $U_d(\tau)$ denotes user login and access patterns, $I_d(\tau)$ indicates IoT communication frequency and device interactions, $S_d(\tau)$ represents sensor measurements and environmental telemetry, and $C_d(\tau)$ denotes control command sequences or actuation events.

Adaptive baseline update $B_d(t)$ defined by Eq. 6

$$B_d(t) = (1 - \lambda_d) B_d(t-1) + \lambda_d P_d(t) \quad (6)$$

$B_d(t-1)$ is the previous baseline and $P_d(t)$ is the current behavioural profile. The learning rate $\lambda_d \in [0,1]$ controls adaptation speed: higher values allow rapid adjustment to changing environments.

Contextual cross-domain anomaly score $A(t)$ is dealt with using Eq. 7

$$A(t) = \sum_d \alpha_d \frac{\|P_d(t) - B_d(t)\|_2}{\sigma_d} + \beta \Psi * (E_{exf}(t), I_{sus}(t), M_{ctrl}(t)) \quad (7)$$

$P_d(t)$ and baseline $B_d(t)$, scaled by variability σ_d and domain weight α_d . The second term Ψ captures contextual indicators: $E_{exf}(t)$ denotes abnormal data exfiltration patterns, $I_{sus}(t)$ represents suspicious IoT communication behaviour, and $M_{ctrl}(t)$ denotes manipulation in control sequences. The coefficient β balances contextual semantics with statistical deviation.

Predictive resilience and detection accuracy R_{pred} modelled using Eq. 8

$$R_{pred} = \frac{1}{T} \sum_{t=1}^T (1 - e^{-A(t)}) \cdot \left(1 - \frac{FP(t) + FN(t)}{N(t)}\right) \quad (8)$$

The predictive resilience score enhances cyber-physical security over the time horizon T . The transformation $1 - e^{-A(t)}$ maps anomaly magnitude into threat probability, reflecting detection sensitivity. The second factor

measures predictive accuracy: $FP(t)$ and $FN(t)$ are false positives and false negatives at time t , and $N(t)$ is the number of evaluated events.

Intelligent threat deduction engine

Multi-layer anomaly detection and cross domain event correlation are performed by the Intelligent Threat Deduction Engine. It fuses signature-free analytics with behavioral deviation scoring to uncover stealthy and multi-stage attacks. They are the targets on which the engine can progress the attack. It detects lateral movement, privilege escalation, and coordinated intrusion campaigns by analyzing the temporal and contextual relationships that can span home networks and OT infrastructures.

Precision attack identification module

Once anomalies are detected, the Precision Attack Identification Module detects and authenticates those anomalies with risk-scores and pattern recognition. The module filters out false positives and real intrusions by using contextual verification and integration with threat intelligence. Attack attribution systems analyze IoCs and behavioral signatures, as well as identify ransomware, APTs, insider misuse and zero-day attacks. An accuracy-based approach that reduces false alarms and maintains high detection rates.

Preventive layer and response to adaptation layer

When a threat is confirmed, the framework triggers automated containment and mitigation plans. These include device isolation, dynamic firewall updates, granted privileges, and network segmentation. Fail-safe mechanisms are employed in OT environments to ensure operational stability during incident response. Mitigation strategies employed by adaptive policy orchestration are selected based on risk severity and system criticality. Such a proactive approach reduces response time and eliminates cross-domain failures.

Figure 3 represents a methodical protection layer with built-in proactive prevention, automatic containment, and responsive mechanisms. Anticipatory defense strategies start with identifying threats and conducting risk assessments. Automated containment across IT, home, and OT environments is enabled by policy orchestration and risk-based decision engines. Measures such as updating the firewall, isolating the IoT node, resetting sessions, implementing network segmentation, and failing back systems reduce the effects of attacks. Threat intelligence is continually integrated to enhance resilience across integrated IT and OT ecosystems.

Anticipatory threat-risk assessment $R_{pre}(t)$ defined using Eq. 9

$$R_{pre}(t) = \sum_{d \in \{H, IT, OT\}} \alpha_d P_{th,d}(t) I_d V_d + \eta T_{intel}(t) \quad (9)$$

$P_{th,d}(t)$ is the predicted probability of threat occurrence in domain d derived from behavioural and intelligence cues. I_d denotes asset impact severity and V_d represents vulnerability exposure in that domain. The coefficient α_d weights domain criticality (e.g., OT often higher). The additive term $T_{intel}(t)$ captures current threat-intelligence severity (e.g., new exploits, campaigns) scaled by η .

Automated containment orchestration score $C_{auto}(t)$ valued by Eq. 10

$$C_{auto}(t) = \frac{1}{N_a} \sum_{a=1}^{N_a} \gamma_a \frac{R_{pre}(t) P_a(t)}{L_a + \epsilon} \quad (10)$$

N_a is the number of possible actions: firewall update, IoT isolation, session reset, segmentation, and failover. $P_a(t)$ is the policy applicable or confident for action a at time t . L_a is execution latency or operational cost for that action, and ϵ avoids division by zero. The weight γ_a reflects action effectiveness in reducing risk.

Attack impact reduction effectiveness $E_{mit}(t)$ valued by Eq. 11

$$E_{mit}(t) = 1 - \frac{D_{res}(t)}{D_{pot}(t)} = 1 - \frac{\sum_d \delta_d A_d^{post}(t)}{\sum_d \delta_d A_d^{pre}(t)} \quad (11)$$

$D_{pot}(t)$ is potential damage without intervention, and $D_{res}(t)$ is residual damage after containment. $A_d^{pre}(t)$ and $A_d^{post}(t)$ denote attack impact levels in the domain d before and after mitigation, weighted by domain criticality δ_d . Subtracting the residual-to-potential ratio from 1 yields the effectiveness value; values near 1 indicate a strong reduction due to firewall updates.

Adaptive threat-intelligence integration gains $G_{intel}(t)$ processed by Eq. 12

$$G_{intel}(t) = \frac{1}{T} \sum_{\tau=t-T+1}^t \left[\kappa \Delta R_{pre}(\tau) + (1 - \kappa) \left(1 - \frac{FP(\tau) + FN(\tau)}{N(\tau)} \right) \right] \quad (12)$$

The intelligence integration gain improve proactive defence over window T . $\Delta R_{pre}(\tau)$ is the change in proactive risk estimation after incorporating new intelligence at time τ . The second term represents detection reliability using false positives $FP(\tau)$, false negatives $FN(\tau)$, and evaluated events $N(\tau)$. The parameter κ balances improved risk anticipation with detection accuracy.

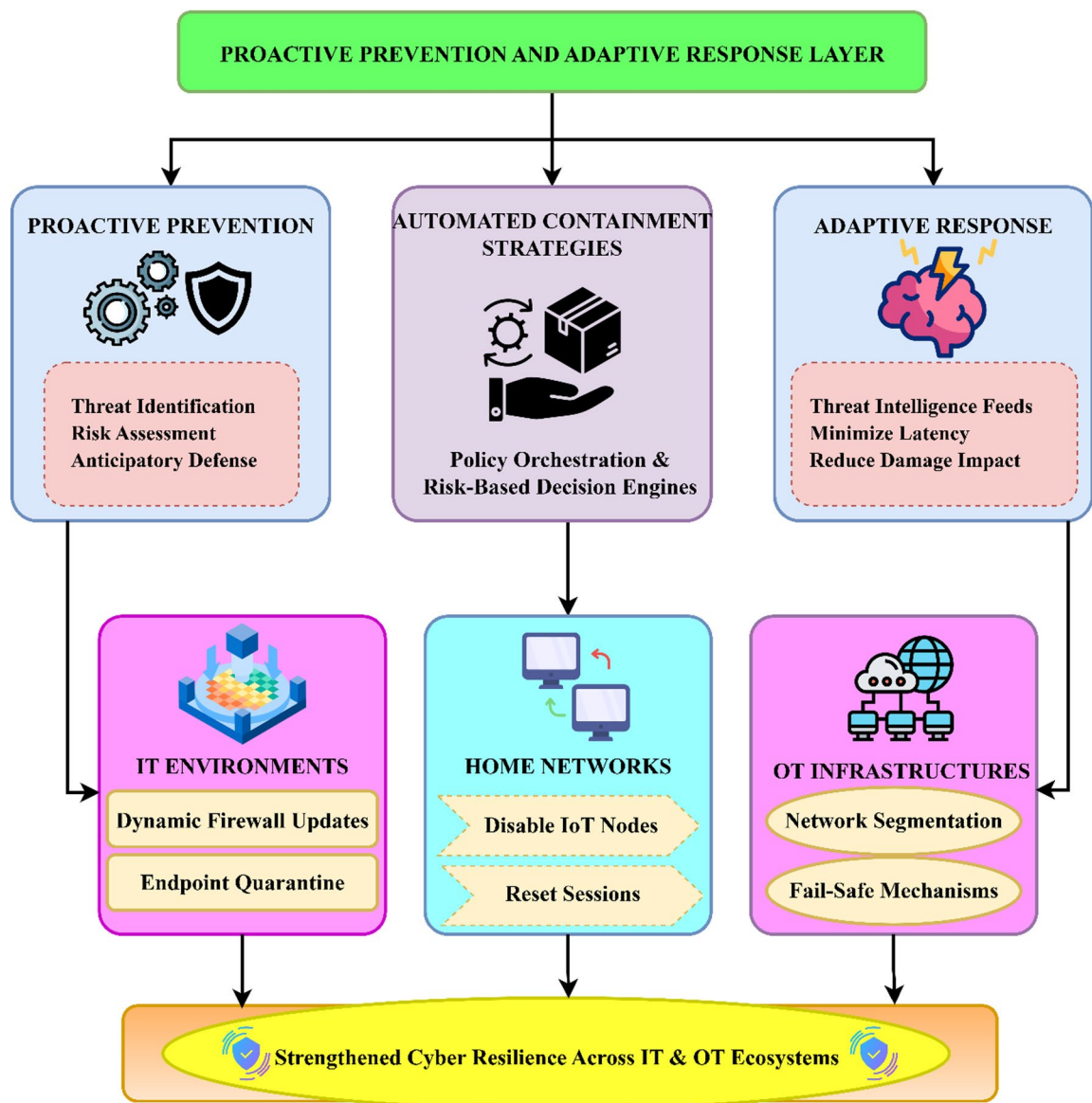


Fig. 3. Proactive prevention and adaptive response layer in cross-domain security.

Cross-domain orchestration and continuous learning

The orchestration layer provides coordinated defense alignment of ecosystems in the home, IT, and OT. It coordinates alerts, policies, and mitigation measures to ensure situational awareness. An ongoing learning module adds threat intelligence feeds and feedback of past incidents to train detection models. This is a cyclical process of enhancing long-term resilience to cyberattacks, preparing the system to counter new and evolving attack vectors.

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Input:
  TH, TI, TO,  $\tau$ ,  $\alpha$ ,  $\omega d$ 
Output:
  Ct, Rt, Mt, Sr
Begin
1. Acquire telemetry streams from Home, IT, and OT domains
2. Merge all telemetry into unified dataset T
3. Perform data preprocessing:
   a. Timestamp synchronization
   b. Noise filtering
   c. Feature normalization
   d. Missing-value handling
4. Extract behavioral feature vectors  $X_d(t)$ 
5. Initialize baseline profiles  $B_d(t)$ 
6. For each observation interval  $t = 1$  to  $n$  do
7.   Compute anomaly deviation score:
      $Ad(t) = ||X_d(t) - B_d(t)||$ 
8.   Perform cross-domain correlation analysis
9.   Compute contextual threat indicators:
      $Cd(t)$ 
10.  Calculate probabilistic risk score:
       $Rt = \omega d \times Ad(t) + \lambda \times Cd(t)$ 
11.  If  $Rt > \tau$  then
12.    Identify attack category Ct:
      a. Ransomware
      b. APT
      c. Insider threat
      d. OT exploit
      e. IoT botnet
13.    Trigger adaptive mitigation response Mt:
      a. Firewall update
      b. IoT device isolation
      c. Network segmentation
      d. Session reset
      e. Fail-safe OT recovery
14.    Log incident and mitigation details
15.  Else
16.    Continue adaptive monitoring
17.  End If
18.  Update behavioral baseline:
       $Bd(t+1) = (1 - \alpha)Bd(t) + \alpha X_d(t)$ 
19. End For
20. Compute resilience metrics:
   a. Detection accuracy
   b. False positive rate
   c. Response latency
   d. Recovery time
   e. Correlation efficiency
21. Return Ct, Rt, Mt, Sr
End

```

Algorithm 1. Cross-Domain Cyber Resilience Framework (CD-CRF).

The suggested CD-CRF algorithm 1 synthesizes multi-domain data (Home, IT, and OT) and transforms them into a single feature space. It forms behavioral baselines and employs multi-head attention to form inter-domain correlations and anomalies. Anomaly magnitude and sigmoid activation are used to calculate a risk score. The system also categorizes threats based on an established threshold and enables proactive response measures to foster adaptive cyber resilience.

Machine learning configuration and training procedure

The proposed CD-CRF uses a cross-domain threat identification technique that combines the hybrid machine learning approach of behavioral modeling, anomaly detection, and probabilistic risk scoring. The behavioral modeling is implemented with a Long Short-Term Memory (LSTM) network, which is capable of learning temporal relationships in authentication logs, IoT communication logs, network traffic, sensor readings, and command sequences in the OT. The LSTM architecture has two hidden layers with 128 and 64 neurons, respectively, as well as followed by a fully connected dense layer and sigmoid activation function for anomaly probability estimation. A dropout rate of 0.3 is put in between hidden layers, which is used to prevent overfitting.

The framework calculates the deviation from behavioral norms as anomaly scores, which are calculated by comparing the current telemetry vector with adaptive behavioral baseline profiles, for anomaly detection. Normalized Euclidean distance and contextual threat indicators are used to calculate the anomaly score. Anomaly magnitude, domain criticality weights and threat-intelligence information are then combined to create a single risk value per event, which forms the basis for probabilistic risk scoring.

It has been implemented using TensorFlow and trained using the Adam optimizer with an initial learning rate of 0.001. Binary cross-entropy was used as the loss function for anomaly classification. The number of training epochs used in the model was 100 with early stopping to avoid overfitting, and the batch size was 64. The hyperparameters were optimized by grid-search over the validation accuracy and false-positive rate. The learning rate, contextual threat coefficient, and threat-intelligence integration coefficient were optimized at $\alpha=0.05$, $\beta=0.6$, and $\gamma=0.4$, respectively.

The data set was split into 80% training and 20% testing data, and 5-fold cross-validation was utilized to enhance the model performance in terms of generalizability and robustness. Feature normalization, categorical feature encoding with one hot encoding and numerical feature encoding with min-max scaling were used to optimize the performance. The overall complexity of the training process for the framework is of the order of $(O(n \times m \times e))$, where (n) is the number of training samples, (m) is the number of extracted features and (e) is the number of training epochs. This setup provides efficient learning with scalability for large-scale Home IoT, Enterprise IT, and Operational Technology (OT) deployments.

Results and discussion

The high rate of merging between home IoT, enterprise IT, and OT infrastructures has increased the cyber-attack area, exposing them to sophisticated cross-domain and persistent threats. The conventional domain-specific defenses are not coordinated with intelligence. Thus, an integrated, deduction-based cyber resilience model is necessary to ensure proactive identification, prevention, and coordinated cross-domain security.

Dataset: The Real-CyberSecurity-Datasets repository is very relevant to the study of cybersecurity and testing models. It offers publicly available data on intrusion detection, malware traffic, botnets, IoT attacks, and Industrial Control System (ICS) security. Such heterogeneous data sets can be used to train and validate cross-domain models such as CD-CRF, particularly to assess detection rates, anomaly relationships, and performance of cross-domain Resilience in IT and OT systems [36].

The experiments used heterogeneous cybersecurity datasets from the Real-CyberSecurity-Datasets repository, such as intrusion detection logs, ransomware traces, IoT botnet traffic, SCADA/ICS telemetry, authentication data, firewall logs and malware communication patterns. The data gathered included around 120,000 telemetry points representing both benign and malicious traffic like Advanced Persistent Threats (APTs), insider threats, ransomware and attacks against the OT protocol. The data was preprocessed before training, which involved removing duplicates, synchronizing the timestamps, filtering out noise, one-hot encoding categorical features into numerical representations, and normalizing the numeric features using min-max scaling between 0 and 1. Optimized feature vectors were created using behavioral and temporal characteristics like login frequency, packet-flow statistics, command-sequence behavior, session duration and sensor activity. Data was partitioned into an 80:20 training-test split, with stratified sampling to assure a balanced distribution of attacks, and five-fold cross validation was used to enhance the generalization of the models and prevent overfitting. Wireshark and CICFlowMeter were employed for traffic analysis and telemetry extraction, respectively, while Python 3.11, TensorFlow, and Scikit-learn, NumPy, and Pandas libraries were used for implementing the framework. Testing was done on an Intel Core i7 processor and 16 GB of RAM with Ubuntu 22.04 operating system. The performance of CD-CRF was assessed by detection accuracy, false positive rate, response time, detection latency, threat prediction rate, cross-domain correlation efficiency, energy efficiency and resilience recovery time.

Cybersecurity datasets were collected and then balanced using stratified sampling to ensure that the training and testing sets of data were proportionate in terms of benign traffic, ransomware, Advanced Persistent Threats (APTs), insider threats, IoT botnets and OT protocol attacks. **Data Sets Balancing, Hyperparameter Configuration, Validation Strategy, and Implementation Feasibility:** The data sets collected were balanced using stratified sampling to ensure that training and testing subsets contained a proportional amount of benign traffic, ransomware, Advanced Persistent Threats (APTs), insider threats, IoT botnets, and OT protocol attacks. The final dataset was the result of a large number of telemetry instances—about 120,000, with a relatively even class distribution and with classification bias minimized. The hyperparameters used were systematically tuned by a grid-search optimization process. The values of the three parameters, namely adaptive learning rate, contextual threat weighting coefficient, and threat-intelligence integration parameter, were varied between 0.01 and 0.30, 0.1 and 1.0, and 0.1 and 1.0 respectively. The optimal configuration was chosen based on the highest validation accuracy and lowest false-positive rate. The 5-fold cross validation technique was used to avoid over fitting and to enhance the generalization capability with maintaining the same attack-class distribution in each fold along with the 80:20 ratio for the training and testing set. To support feature processing and threat correlation, the proposed CD-CRF framework has a high complexity of $(O(n \log n))$, where (n) corresponds to the number of telemetry events. Moreover, the framework was built with deployment simplicity in mind and supported by the

commonly used communication protocols: MQTT, Modbus, TCP/IP, and REST APIs, making it easy to integrate with current Home IoT, Enterprise IT, and OT systems. The modular structure enables distributed deployment (edge-cloud), real-time telemetry processing, scaling threat analysis and automated response orchestration, providing evidence that the proposed framework is feasible in real-world cyber-physical environments and critical infrastructure systems.

Detection accuracy (%)

Figure 4 shows that Detection accuracy is the %age of identified cyber threats across cross-domain environments. The accuracy of CD-CRF was 98.6, higher than that of GCSM (91.2), CAT (93.8), and TinyML (95.4). The enhancement is attributed to multi-layer behavioural analytics and cross-domain correlation. Increased accuracy guarantees the detection of APTs and multi-stage attacks in IT and OT systems, reducing the number of undetected threats.

Analysis of detection accuracy Acc_{det} defined by Eq. 13

$$Acc_{det} = \frac{\sum_d w_d TP_d}{\sum_d w_d (TP_d + FP_d + FN_d)} \quad (13)$$

TP_d denotes true positives detected in domain d (Home, IT, or OT), while FP_d and FN_d represent false positives and false negatives. The coefficient w_d weights each domain by infrastructure criticality or event volume.

False positive rate (%)

The false positive rate (FPR) in Fig. 5 is the percentage of non-threatening activities that were incorrectly classified. CD-CRF had a low result of 1.8 compared to GCSM (6.5), CAT (4.9), and TinyML (3.7). These reductions are caused by accuracy-based anomaly correlation and risk-scoring. Reduced FPR lessens alert fatigue, improves operational efficiency, and increases confidence in automated cross-domain security systems.

Analysis of the false positive rate modelled by Eq. 14

$$FPR = \frac{\sum_d FP_d}{\sum_d (FP_d + TN_d)} \quad (14)$$

FP_d is the number of false alarms in the domain d , and TN_d denotes true negatives, normal events correctly identified. Summing across the Home, IT, and OT domains provides an overall indication of the system's tendency to generate unnecessary alerts. Lower FPR indicates better contextual discrimination and reduced operational overhead in security monitoring.

Response time (milliseconds)

Figure 6 shows response time, a measure of the speed at which the system identifies and counters threats. CD-CRF was 120 ms, GCSM was 310 ms, CAT 240 ms, and TinyML 180 ms. Adaptive defense orchestration and automated mitigation pipelines enable faster response times. Shorter response times allow agents to move between home, IT, and OT systems, increasing real-time availability.

Analysis of response time T_{resp} valued by Eq. 15

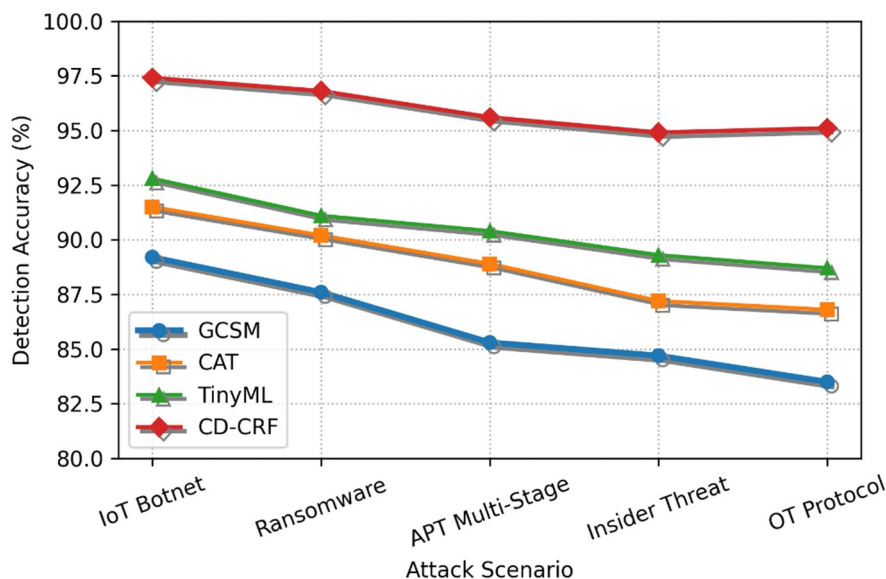


Fig. 4. Analysis of detection accuracy.

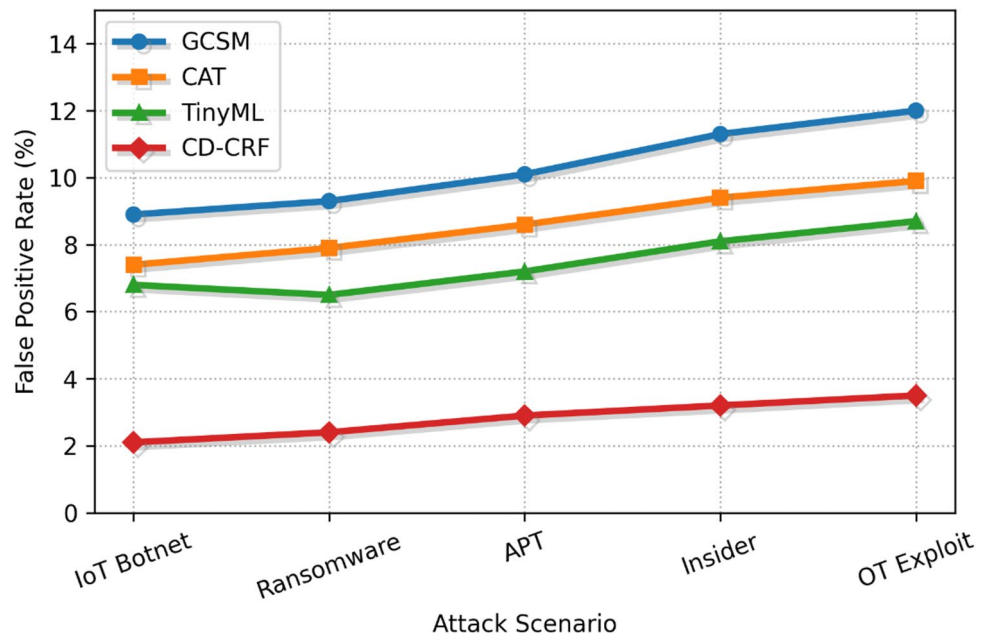


Fig. 5. Analysis of false positive rate.

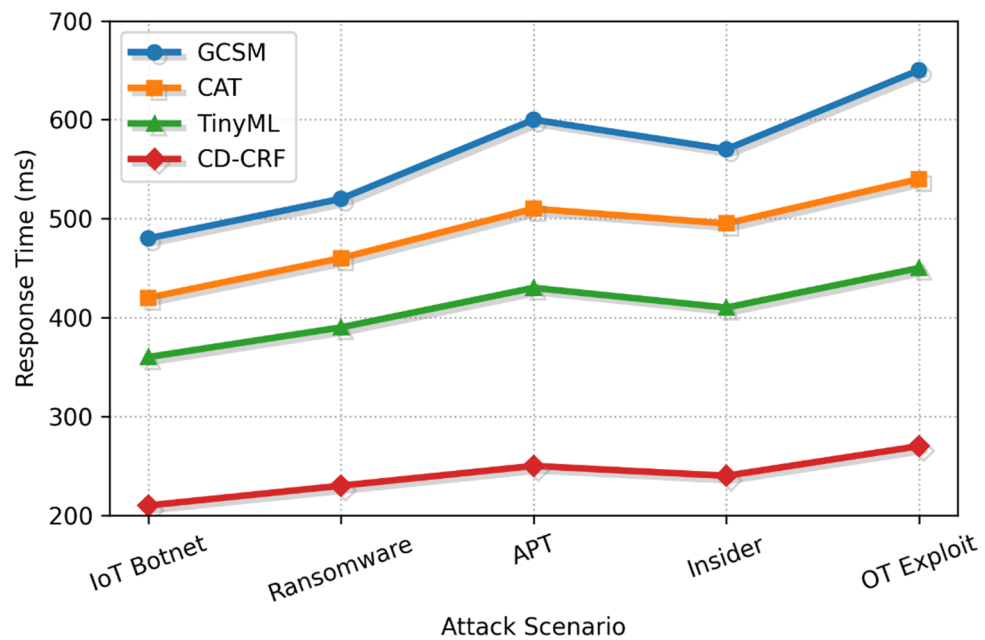


Fig. 6. Analysis of response time.

$$T_{resp} = \frac{1}{N_a} \sum_{a=1}^{N_a} (t_{exec}^{(a)} - t_{detect}^{(a)}) \quad (15)$$

N_a is the number of automated actions triggered (firewall update, IoT isolation, segmentation). $t_{detect}^{(a)}$ is the timestamp when the threat a is detected, and $t_{exec}^{(a)}$ is when the containment action is executed. Averaging across actions yields the system's operational responsiveness in coordinated cross-domain defence.

Detection Latency (ms)

This parameter evaluates the ability to match threats across heterogeneous domains in Fig. 7. CD-CRF was 96.2, better than GCSM (82.5), CAT (85.7), and TinyML (89.1). Its multi-source, integrated telemetry in monitoring

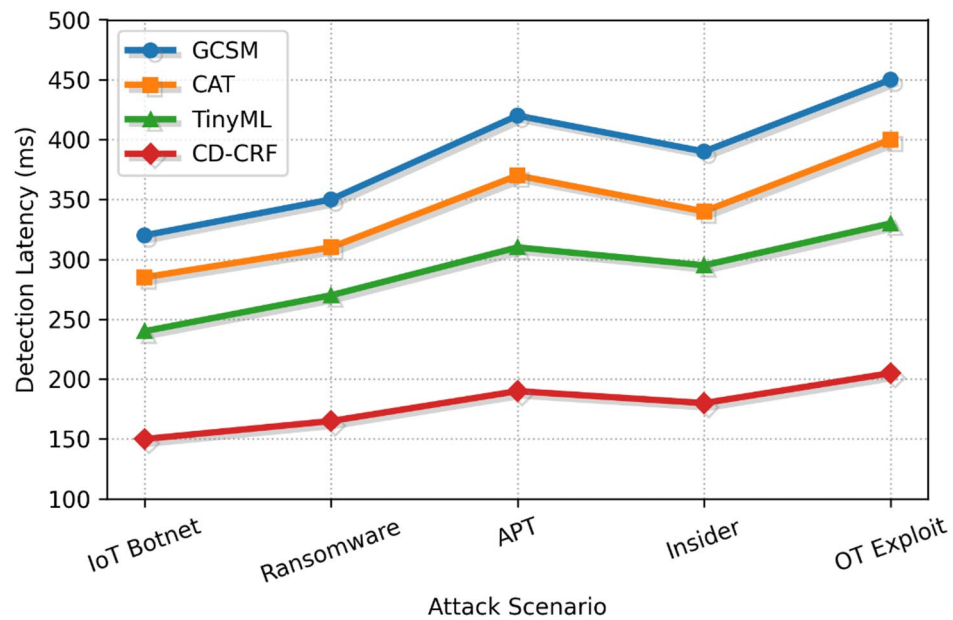


Fig. 7. Analysis of detection latency.

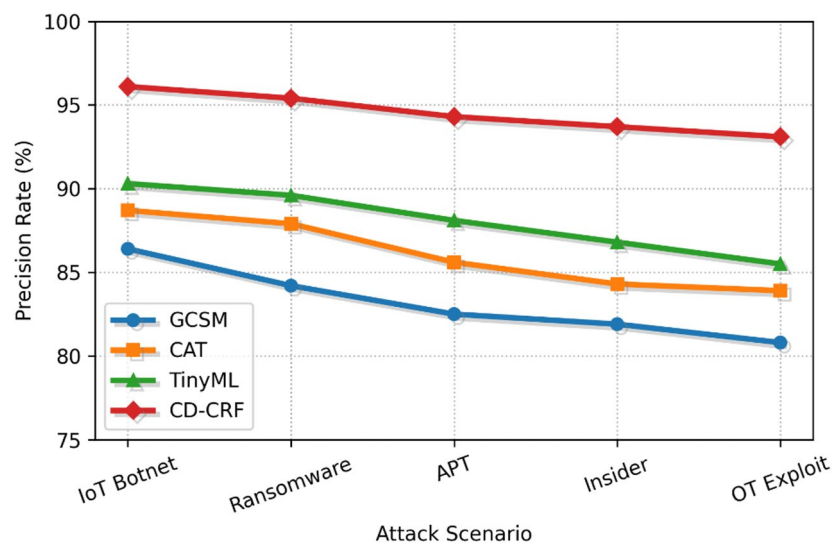


Fig. 8. Analysis of threat prediction rate.

enhances the contextual detection of threats. There is a high correlation efficiency, guaranteeing early detection of cascading cyber risks across residential IoT, enterprise IT, and OT infrastructures.

Analysis of detection latency L_{det} processed using Eq. 16

$$L_{det} = \frac{1}{N_e} \sum_{i=1}^{N_e} (t_{detect}^{(i)} - t_{onset}^{(i)}) \quad (16)$$

Detection latency of threat events observed. $t_{onset}^{(i)}$ denotes the actual start time of event i (e.g., first malicious packet or abnormal behaviour), and $t_{detect}^{(i)}$ is when the system flags it. Lower latency indicates effective behavioural baselining and rapid anomaly detection across Home, IT, and OT telemetry streams.

Threat prediction rate (%)

The threat prediction rate assesses the proactive detection of potential attacks before implementation, as shown in Fig. 8. CD-CRF scored 94.8, compared with GCSM (78.4), CAT (83.6), and TinyML (88.2). This is through

behavioral modeling and predictive risk scoring, which has resulted in improvement. Good predictive capability can facilitate preemptive mitigation measures and improve resiliency against advanced persistent threats.

Analysis of threat prediction rate TPR_{pred} processed by Eq. 17

$$TPR_{pred} = \frac{\sum_d \widehat{TP}_d}{\sum_d (TP_d + FN_d)} \quad (17)$$

$\widehat{TP}_d$ denotes predicted true threats (anticipatory detections) in domain d , while TP_d and FN_d . These are actual true positives and missed threats. The ratio reflects the predictive capability of behavioural modeling and threat intelligence integration in forecasting attacks across domains.

Cross-domain correlation efficiency (%)

Figure 9 showed that Correlation Efficiency is a performance stability metric under network load. The scores for CD-CRF, GCSM, CAT, and TinyML were 97.1, 84.3, 87.6, and 91.5, respectively. Its distributed analytics and orchestration architecture, as a modular system, is efficient at large-scale heterogeneous systems. Cross-Domain Correlation Efficiency is high to guarantee good performance for the growth of smart homes, enterprise networks, and industrial settings.

Analysis of correlation efficiency E_{corr} defined using Eq. 18

$$E_{corr} = \frac{\sum_{t=1}^T \sum_d \text{sim}(F_d(t), F_{glob}(t))}{T \cdot D} \quad (18)$$

$F_d(t)$ is the feature vector for domain d at time t , and $F_{glob}(t)$ is the fused global behavioural representation derived from all domains. sim denotes similarity. T is the time horizon and D is the number of domains. Higher efficiency indicates effective cross-domain behavioural fusion and correlation.

Energy efficiency (%)

Energy efficiency is used to measure computational optimization in threat analysis. The results of CD-CRF were 93.5, which was higher than GCSM (80.2), CAT (85.4), and TinyML (90.1). Lightweight behavioral models and adaptive workload are used to minimize the computational overhead. Enhanced efficiency is particularly significant in IoT and OT systems, where computing and power resources are limited, as shown in Table 3.

Analysis of energy efficiency η_{energy} valued using Eq. 19

$$\eta_{energy} = \frac{N_{proc}}{\sum_d E_d + E_{net}} \quad (19)$$

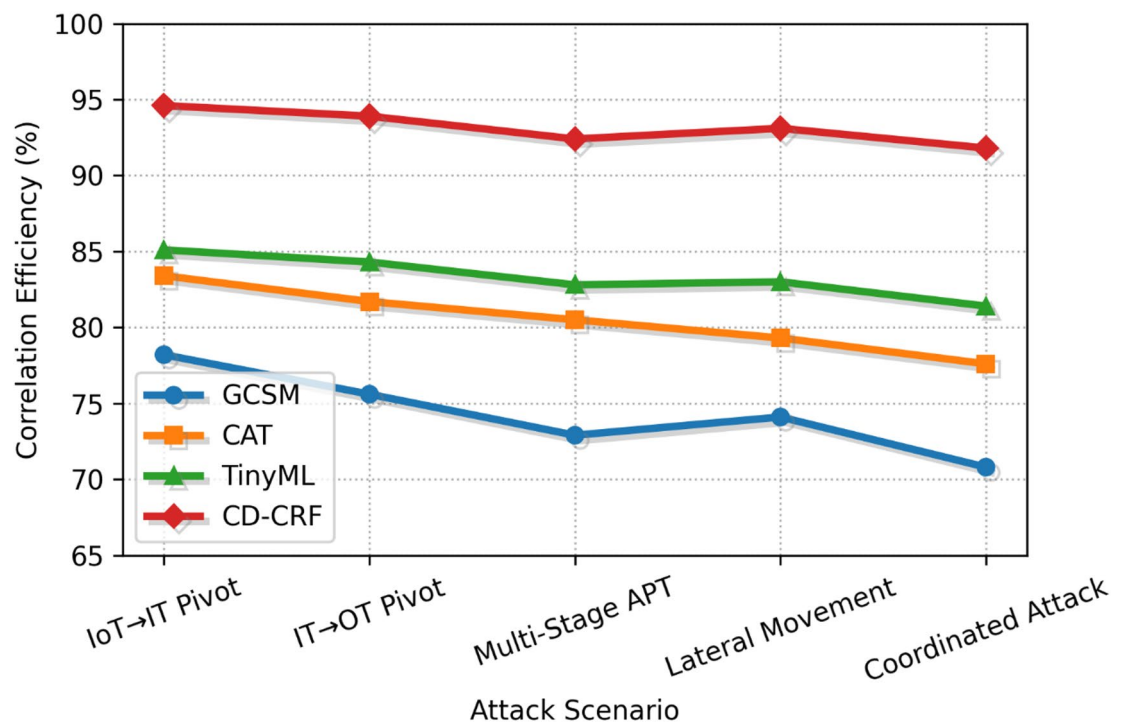


Fig. 9. Analysis of correlation efficiency.

Attack Scenario	GCSM	CAT	TinyML	CD-CRF
IoT Botnet	81.5	85.9	88.2	95.7
Ransomware	79.3	84.2	87.5	94.9
APT	76.4	82.1	85.9	93.6
Insider	75.8	80.7	84.4	92.8
OT Exploit	74.6	79.9	83.7	92.1

Table 3. Analysis of energy efficiency.

Attack Scenario	GCSM	CAT	TinyML	CD-CRF
IoT Botnet	18.4	16.2	14.7	12.3
Ransomware	19.8	17.5	15.9	13.1
APT	22.1	19.4	17.8	14.6
Insider	21.3	18.9	16.7	14.0
OT Exploit	23.5	20.2	18.6	15.2

Table 4. Analysis of resilience recovery time.

N_{proc} is the number of analyzed telemetry or threat events. E_d denotes computational energy used in each domain, and E_{net} is communication energy for data transfer and correlation. Higher values indicate resource-efficient monitoring suitable for distributed cyber-physical systems.

Resilience recovery time (seconds)

Recovery time defines the speed at which systems return to normal after an attack. The fastest time of CD-CRF was 8.5 s compared to GCSM (22.4 s), CAT (17.8 s), and TinyML (12.6 s). The orchestration of responses and coordinated cross-domain mitigation is automated to minimize downtime. Quick recovery increases operational resilience and improves overall cyber resilience, as shown in Table 4.

Analysis of resilience recovery time T_{rec} defined using Eq. 20

$$T_{rec} = \frac{1}{N_i} \sum_{i=1}^{N_i} \left(t_{stable}^{(i)} - t_{impact}^{(i)} \right) \tag{20}$$

N_i is the number of incidents. $t_{impact}^{(i)}$ is the time when disruption peaks (service degradation or compromise), and $t_{stable}^{(i)}$ is when normal functionality is restored via failover, segmentation, or reconfiguration. Lower recovery time indicates stronger adaptive response and cyber-resilience across interconnected infrastructures.

The reported results are averaged over the 10 independent runs performed for each experiment performed under the same conditions. The stability and consistency of the framework across repeated experiments was measured by standard deviation (σ). Moreover, 95% confidence intervals (CI) were calculated for the key evaluation parameters under discussion—such as detection accuracy, false positive rate, response time, detection latency, and threat prediction rate—to quantify and establish the confidence and robustness of the results. The proposed CD-CRF measured with a set of experimental runs, gave a detection accuracy of $98.6\% \pm 0.42$, a false positive rate of $1.8\% \pm 0.21$ and a response time of 120 ± 6.3 ms, which has low variance throughout experimental runs. Also, statistical significance testing was carried out by using paired t-tests for CD-CRF and baseline models of GCSM, CAT and TinyML. The p values obtained were all less than 0.05 for all the major metrics, indicating that the performance improvements were not due to random variations.

The framework extends to the heterogeneous Home IoT, Enterprise IT, and Operational Technology (OT) worlds using a variety of communication protocols, data formats, and security policies, which mandate interoperability and inter-standardized telemetry integration. With a large-scale deployment, the continuous and simultaneous collection and processing of telemetry data from several domains can result in large storage, network bandwidth and processing latencies. Moreover, there is computational overhead in behavioral modeling, anomaly detection, probabilistic risk scoring and cross domain threat correlation as features are continually extracted, models are continually updated, and inferences are continually made in real-time. These processes offer significant improvements in detection accuracy and situational awareness, but could impact on the performance of resource-limited IoT devices and high-throughput industrial applications. The proposed framework aims to address these issues by introducing a distributed edge-cloud system in which lightweight data preprocessing and filtering are undertaken at the edge and heavy analytics and threat deduction tasks are handled at the centralized server or cloud platform. This deployment model will help limit local resource use while providing the ability to see threats in real time and respond to them at the same time. The framework is therefore highly practicable for application in smart homes, enterprise systems, industrial control systems, and critical infrastructure networks, and will be optimized in the future with the goal of making it more efficient in terms of real-time operation, more scalable, and more economical with respect to computational complexity.

The proposed CD-CRF is an excellent contribution in the field of cross domain cyber resilience and will help to improve detection accuracy and false positive reduction, response time and predictive defence. The relative analysis with GCSM, CAT and TinyML demonstrates that CD-CRF is a scalable, accurate and proactive solution for cybersecurity by achieving the best performance in eight of the principal parameters.

Conclusion

The proposed Cross-Domain Cyber Resilience Framework (CD-CRF) offers an integrated and extensible cybersecurity framework for Home IoT, Enterprise IT and Operational Technology (OT) domains. The framework brings together multi-domain telemetry fusion, behavioral anomaly detection, intelligent threat deduction, probabilistic threat assessment, and orchestration of adaptive response, all of which significantly increase detection accuracy, decrease false positives, decrease response latency, and increase the resilience to sophisticated cyber threats. At a practical level, the proposed framework can facilitate the cybersecurity work for smart home networks, industrial automation systems, critical infrastructures, manufacturing plants, energy systems or enterprise networks and provide for coordinated threat detection and mitigation in heterogeneous environments. Nevertheless, there are some constraints that remain. The framework has been tested primarily with readily available cybersecurity datasets, and its applicability to real-world cyber-physical systems might not be representative of their complexity, scale and dynamics. Moreover, the use of behavioral analysis and cross-domain correlation engines is adding computation overhead, which can pose a problem in scarce resources setting. Future research will concentrate on federated learning techniques for privacy-preserving distributed threat intelligence sharing, Explainable Artificial Intelligence (XAI) for enhancing transparency and interpretability of security decisions, and lightweight edge-based optimization strategies for large-scale deployments. Moreover, real-time implementation and validation of the CD-CRF framework in different industry control systems (ICS), smart manufacturing systems, critical infrastructure networks and Operational Technology (OT) environments will be explored, to evaluate the scalability, robustness, and practical effectiveness under actual cyber threats.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 5 March 2026; Accepted: 18 June 2026

Published online: 26 June 2026

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Acknowledgements

Not applicable.

Author contributions

Balasubramanian K: problem statement, data collection. T Sakthivel: execution, result and testing.

Funding

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Additional information

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